

Multiple Choice (10 marks)

1. Let $f(X) = \text{if } x = 1 \text{ then } 0 \text{ else } [x * f(x - 1) + x**2]$. The value of $f(4)$ is
 - (a) 53
 - (b) 29
 - (c) 50
 - (d) 100
2. Local caching of files is common in distributed file systems, but it has the disadvantage that
 - (a) Temporary inconsistencies among views of a file by different machines can result
 - (b) The file system is likely to be corrupted when a computer crashes
 - (c) A much higher amount of network traffic results
 - (d) Caching makes file migration impossible
3. Which is the smallest asymptotically?
 - (a) $O(1)$
 - (b) $O(n)$
 - (c) $O(n^2)$
 - (d) $O(\log n)$
4. Which of the following language features requires that stack-based storage allocation be used rather than static allocation?
 - (a) Recursive procedures
 - (b) Arbitrary goto's
 - (c) Two-dimensional arrays
 - (d) Integer-valued functions
5. A 3-way, set-associative cache is
 - (a) one in which each main memory word can be stored at any of 3 cache locations
 - (b) effective only if 3 or fewer processes are running alternately on the processor
 - (c) possible only with write-back
 - (d) faster to access than a direct-mapped cache

True / False (10 marks)

Answer True or False

6. True or False: The upper bound for the value of $f(N)$ is $O(N \log N)$, which suggests a linear growth rate with logarithmic factors.
7. True or False: An Internet Protocol (IP) address is assigned to a device when it is connected to the Internet.
8. True or False: The maximum value of the list $l = [1,2,3,4]$ in Python 3 is 4.
9. True or False: Not all line segments in bitmap graphics can be displayed as perfectly straight due to pixelation and resolution limitations.
10. True or False: In Python 3, the expression `['a', 'Chemistry', 0, 1][1]` evaluates to 'Chemistry'.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to check whether the given string is made up of two alternating characters or not.
12. Write a function to move all the numbers in it to the given string.

End of Paper